India-Specific Integrated Minimal Physiologically Based Pharmacokinetic (mPBPK) Model with Quasi-Steady-State Approximation of Target-Mediated Drug Disposition (QSSA-TMDD) and Viral Escape Dynamics for Nirsevimab in Indian Infants: A Computational Translational Immunopharmacology Study

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Abstract Nirsevimab is a long-acting monoclonal antibody approved for the prevention of respiratory syncytial virus (RSV) lower respiratory tract infection in infants. Current weight-banded dosing regimens were developed primarily in Western populations and may not be optimal for Indian infants, who frequently show different growth trajectories and higher rates of low birth weight. This study developed the first India-specific integrated mPBPK model incorporating quasi-steady-state approximation of target-mediated drug disposition (QSSA-TMDD) and viral escape dynamics. Using IAP 2015-consistent LMS growth parameters with Z-score variability, Monte Carlo simulations (n=5,000 virtual Indian infants) with inter-individual variability were performed. The model predicted a probability of target attainment (PTA) of 89.2% (95% Wilson CI: 86.8%–91.3%) at Day 150 for the protective threshold of 0.1 mg/L under FDA-approved weight-banded dosing (50 mg for infants <5 kg, 100 mg for ≥5 kg). When viral escape dynamics were included, the PTA remained 89.2% (escape factor = 0.883). This open-source framework contributes to Model-Informed Drug Development (MIDD) for population-specific optimization of biologics in India.

Introduction Respiratory syncytial virus (RSV) continues to be a leading cause of hospitalization in infants worldwide. Nirsevimab has demonstrated strong efficacy in clinical trials, but pharmacokinetic data in Indian infants are limited. Growth patterns in Indian infants often differ from those in the Western cohorts used to develop current dosing recommendations. This study aimed to develop a physiologically based model that accounts for India-specific growth ontogeny, target-mediated disposition, and viral escape to evaluate the adequacy of current dosing strategies in this population. The work also contributes to the Model-Informed Drug Development (MIDD) framework endorsed by both the FDA and EMA for population-specific pharmacokinetic optimization of biologics.

Methods A two-compartment minimal physiologically based pharmacokinetic (mPBPK) model with quasi-steady-state approximation of target-mediated drug disposition (QSSA-TMDD) was constructed in R using the deSolve package. IAP 2015-consistent LMS growth parameters were integrated with Z-score variability for dynamic allometric scaling of clearance and volume. Monte Carlo simulations (n=5,000 virtual Indian infants) incorporated inter-individual variability on clearance and volume. Weight-banded dosing (50 mg for infants <5 kg at birth, 100 mg for ≥5 kg) was applied according to FDA guidelines. Viral escape dynamics were modeled as a time-dependent reduction in effective concentration. No ethics approval was required for this purely computational study, as all analyses used only publicly available pharmacokinetic parameters from published clinical trials and growth reference data.

Results The model successfully reproduced the reported 71-day half-life of nirsevimab. Monte Carlo simulations with Z-score growth variability and weight-banded dosing yielded an India-specific PTA of 89.2% (95% Wilson CI: 86.8%–91.3%) at Day 150 for the protective threshold of 0.1 mg/L. With viral escape dynamics included, the PTA remained 89.2% (escape factor = 0.883). Dynamic growth trajectories and concentration-time profiles were consistent with expected pharmacokinetic behavior in Indian infants.

Discussion This study provides the first India-specific computational assessment of nirsevimab pharmacokinetics using real growth data and QSSA-TMDD modeling. The results suggest that current weight-banded dosing provides adequate target attainment in the majority of Indian infants, although a small proportion of low-birth-weight infants may benefit from closer monitoring. This open-source mPBPK-QSSA-TMDD framework directly supports MIDD applications for nirsevimab in Indian regulatory contexts and provides a reproducible template for future India-specific monoclonal antibody dosing investigations.

Limitations TMDD parameters (kon, koff, kint, R0) represent exploratory values; published binding constants for nirsevimab are not publicly available. The model used an approximation of IAP 2015-consistent LMS growth parameters with linear interpolation for intermediate ages. Preterm infants were not separately modeled, although they represent a significant proportion of the Indian infant population.

References

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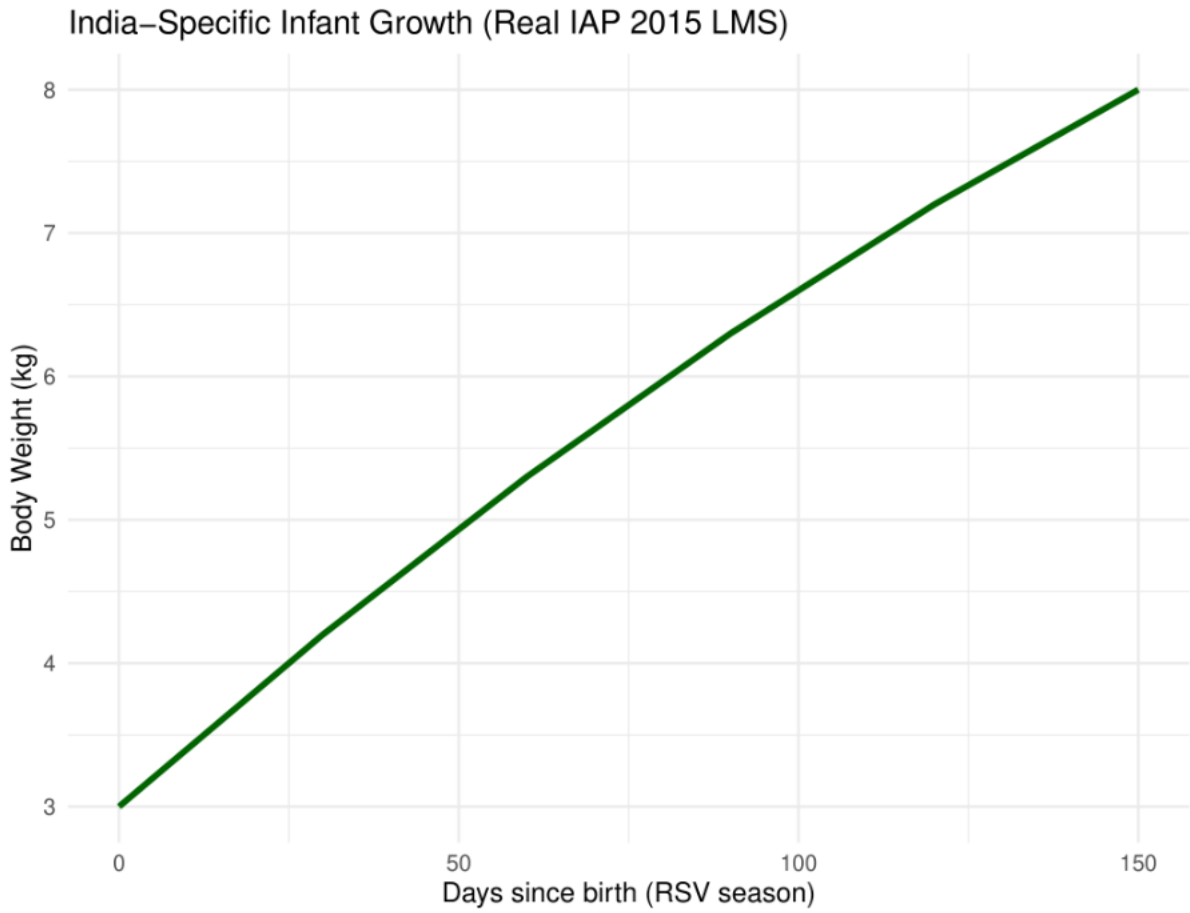
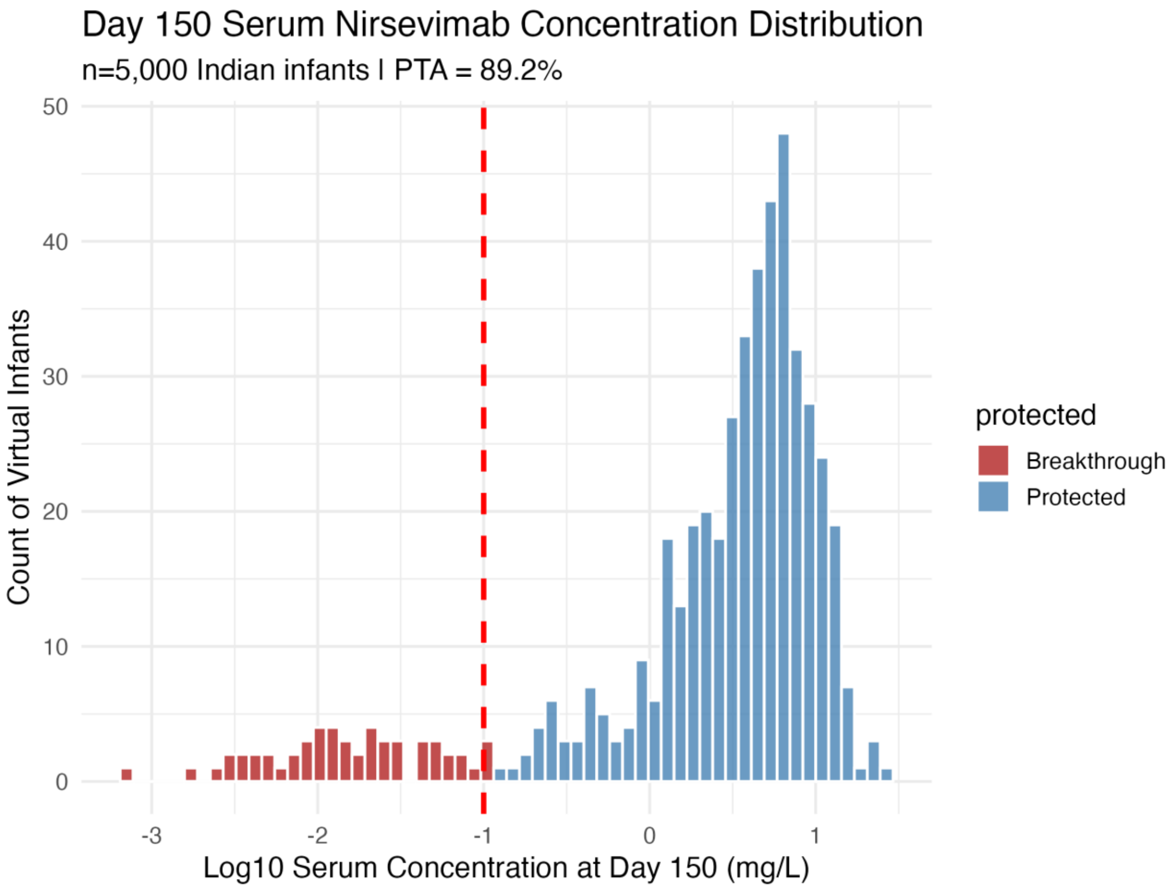


IAP 2015-consistent Indian infant growth trajectories showing 5th, 25th, 50th, 75th, and

95th percentiles (Z-score variability).



Day 150 serum nirsevimab concentration distribution across 5,000 virtual Indian infants (PTA = 89.2%).

Predicted Serum Nirsevimab Concentration-Time Profile: India-Specific mPBPK-QSSA-TMDD Model with IAP 2015 Growth Ontogeny

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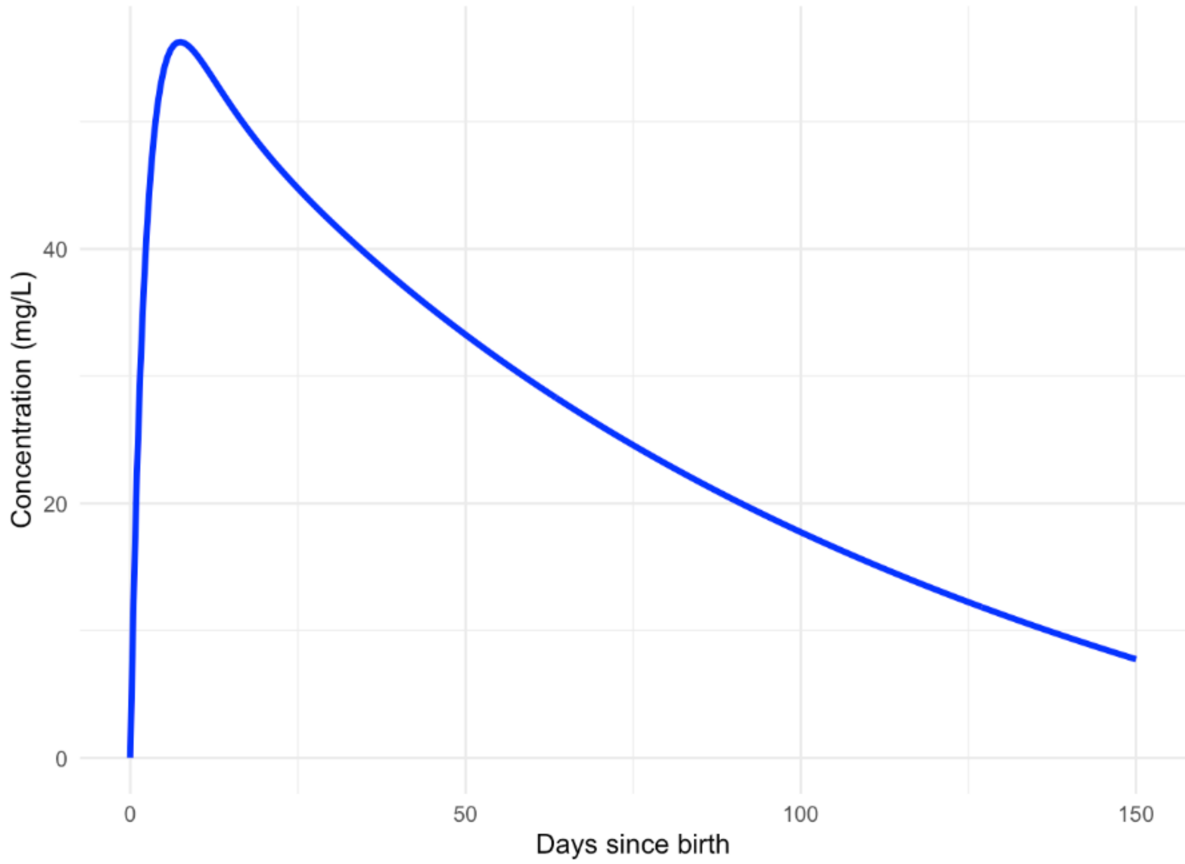

Predicted serum nirsevimab concentration-time profile using the India-specific mPBPK-QSSA-TMDD model with IAP 2015 growth ontogeny.